

Data Science

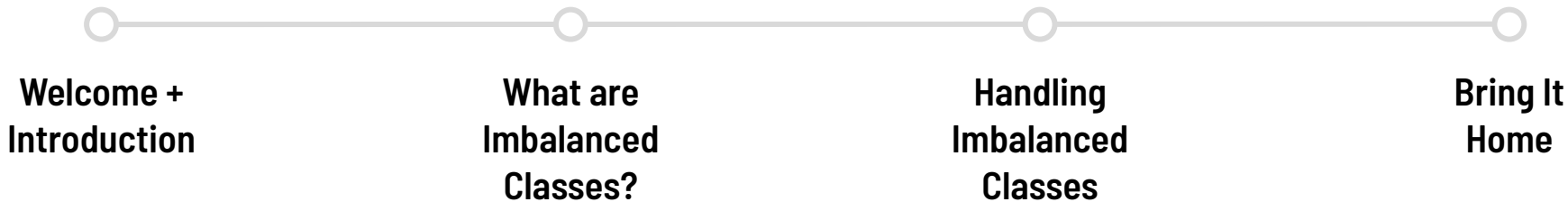
Imbalanced Classes

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Describe when imbalanced data is a problem.

2

Identify methods for handling imbalanced data and appropriately evaluating performance.

3

Apply the `imbalanced-learn` library to oversample minority classes.

4

Apply the `imbalanced-learn` library to create synthetic minority class observations.

What Are Imbalanced Classes?

Imbalanced Classes

When we do not have each class equally represented, we say we have **unbalanced classes** or **imbalanced data**.



Calculate Accuracy

Suppose I want to predict the incidence of West Nile Virus (WNV) in Chicago.

- 99.9% of the observations are "no WNV."
- 0.1% of the observations contain "WNV."

What would the accuracy be if our model predicted all values as "no WNV"?

Calculate Accuracy

Suppose I want to predict the incidence of West Nile Virus (WNV) in Chicago.

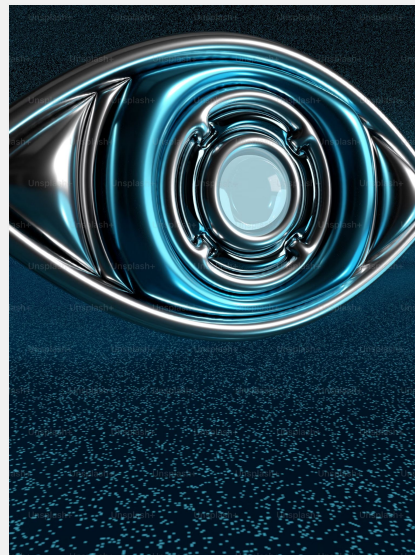
- 99.9% of the observations are "no WNV."
- 0.1% of the observations contain "WNV."

What would the accuracy be if our model predicted all values as "no WNV"?

If we fit a model and tried to optimize for accuracy alone, I can predict "no WNV" for every location and have an **accuracy score of .999!**

Models learn what they see the most

Models struggle to accurately predict data they have seen very few examples of in their training data.



Rare Outcomes

What are examples of business problems where the outcome of interest is very rare?

Rare Outcomes

- Are often very important
- Aren't predicted well by models

So what can you do in this situation?



Handling Imbalanced Classes

Three Key Ways to Improve Your Model

1

Undersampling

2

Oversampling

3

Synthetic data
generation

In general,

**Don't undersample
the majority class.**

You can lose valuable information.

Imbalanced Learn Package

To improve your model's decision function, you should consider using the [imbalanced-learn](#) package to oversample the minority class or create additional synthetic (fake) observations of the minority class.



Imbalanced Classes

Let's see it for ourselves!

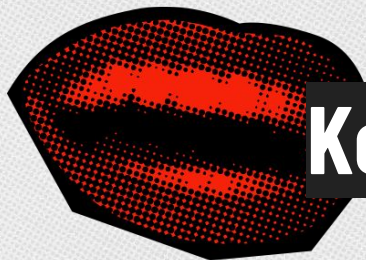
- Open **02-imbalanced-data.ipynb**



Bring It Home

Imbalanced Data

1. Why is imbalanced data (unbalanced target classes) a problem?
2. Is accuracy a good metric if the data are highly imbalanced?
3. How common is imbalanced data?
4. How can we improve an algorithm's performance predicting the minority class?



Key takeaways



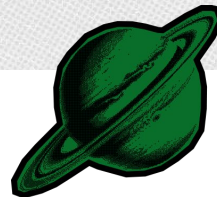
Choose your metric wisely

Metrics can be misleading if you have imbalanced classes



Address imbalances when they matter

Not all imbalances have an impact, but if they do they need to be addressed.



Imbalanced Learn is a helpful tool

Use `make_pipeline` to keep your test set safe.

